3.3 The Rank and the Row Reduced Form

The numbers m and n give the size of a matrix—but not necessarily the *true size* of a linear system. An equation like $0 = 0$ should not count. If there are two identical rows in A, the second one disappears in elimination. Also if row 3 is a combination of rows 1 and 2, then row 3 will become all zeros in the triangular U and the reduced echelon form R. We don't want to count rows of zeros. ***The true size of A is given by its rank***:

DEFINITION ***The rank of A is the number of pivots. This number is r.***

That definition is computational, and I would like to say more about the rank r. The matrix will eventually be reduced to r nonzero rows. Start with a 3 by 4 example.

Four columns
How many pivots?

$$A = \begin{bmatrix} 1 & 1 & 2 & 4 \\ 1 & 2 & 2 & 5 \\ 1 & 3 & 2 & 6 \end{bmatrix}. \tag{1}$$

The first two columns are $(1, 1, 1)$ and $(1, 2, 3)$, going in different directions. Those will be pivot columns. The third column $(2, 2, 2)$ is a multiple of the first. We won't see a pivot in that third column. The fourth column $(4, 5, 6)$ is a combination of the first three (their sum). That column will also be without a pivot.

The fourth column is actually a combination $3(1, 1, 1) + (1, 2, 3)$ of the two pivot columns. ***Every "free column" is a combination of earlier pivot columns.*** It is the *special solutions* s that tell us those combinations of pivot columns:

Column 3 = **2** (column 1)	$s_1 = (-\mathbf{2}, 0, 1, 0)$	$As_1 = \mathbf{0}$
Column 4 = **3** (column 1) + **1** (column 2)	$s_2 = (-\mathbf{3}, -\mathbf{1}, 0, 1)$	$As_2 = \mathbf{0}$

With nice numbers we can see the right combinations. The systematic way to find s is by elimination! This will change the columns but it won't change the combinations, because $Ax = \mathbf{0}$ is equivalent to $Ux = \mathbf{0}$ and also $Rx = \mathbf{0}$. I will go from A to U and then to R:

$$\begin{bmatrix} 1 & 1 & 2 & 4 \\ 1 & 2 & 2 & 5 \\ 1 & 3 & 2 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 2 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 2 & 0 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} \mathbf{1} & 1 & 2 & 4 \\ 0 & \mathbf{1} & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = U$$

U already shows the two pivots in the pivot columns. **The rank of A (and U) is 2.** Continuing to R we see the combinations of pivot columns that produce the free columns:

$$U = \begin{bmatrix} 1 & 1 & 2 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{matrix} \text{Subtract} \\ \longrightarrow \\ \text{row 1} - \text{row 2} \end{matrix} \quad R = \begin{bmatrix} \mathbf{1} & 0 & 2 & 3 \\ 0 & \mathbf{1} & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{2}$$

Clearly the $(3, 1, 0)$ column equals 3 (column 1) + column 2. Moving all columns to the "left side" will reverse signs to -3 and -1, which go in the special solution s:

$$-3 \text{ (column 1)} - \text{(column 2)} + \text{(column 4)} = \mathbf{0} \qquad s = (-3, -1, 0, 1).$$

Rank One

Matrices of ***rank one*** have only *one pivot.* When elimination produces zero in the first column, it produces zero in all the columns. *Every row is a multiple of the pivot row.* At the same time, every column is a multiple of the pivot column!

$$\textbf{Rank one matrix} \qquad A = \begin{bmatrix} \mathbf{1} & 3 & 10 \\ \mathbf{2} & 6 & 20 \\ \mathbf{3} & 9 & 30 \end{bmatrix} \longrightarrow R = \begin{bmatrix} \mathbf{1} & 3 & 10 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

The column space of a rank one matrix is "one-dimensional". Here all columns are on the line through $\boldsymbol{u} = (1, 2, 3)$. The columns of A are $\boldsymbol{u}$ and $3\boldsymbol{u}$ and $10\boldsymbol{u}$. Put those numbers into the row $v^{\mathrm{T}} = [\ 1 \quad 3 \quad 10\ ]$ and you have the special rank one form $A = \boldsymbol{u}v^{\mathrm{T}}$:

$$A = \textbf{column times row} = uv^{\mathrm{T}} \qquad \begin{bmatrix} 1 & 3 & 10 \\ 2 & 6 & 20 \\ 3 & 9 & 30 \end{bmatrix} = \begin{bmatrix} \mathbf{1} \\ \mathbf{2} \\ \mathbf{3} \end{bmatrix} \begin{bmatrix} 1 & 3 & 10 \end{bmatrix}. \tag{3}$$

With rank one, the solutions to $A\boldsymbol{x} = \mathbf{0}$ are easy to understand. That equation $\boldsymbol{u}(v^{\mathrm{T}}\boldsymbol{x}) = \mathbf{0}$ leads us to $v^{\mathrm{T}}\boldsymbol{x} = 0$. All vectors $\boldsymbol{x}$ in the nullspace must be orthogonal to v in the row space. This is the geometry: *row space = line, nullspace = perpendicular plane.* Now describe the special solutions with numbers:

Pivot row [1 3 10]
Pivot variable x_1
Free variables x_2 **and** x_3

$$s_1 = \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix} \qquad s_2 = \begin{bmatrix} -10 \\ 0 \\ 1 \end{bmatrix}$$

The nullspace contains all combinations of s_1 and s_2. This produces the plane $x + 3y + 10z = 0$, perpendicular to the row $(1, 3, 10)$. **Nullspace** (plane) **perpendicular to row space** (line).

Example 1 When all rows are multiples of one pivot row, the rank is $r = 1$:

$$\begin{bmatrix} 1 & 3 & 4 \\ 2 & 6 & 8 \end{bmatrix} \text{ and } \begin{bmatrix} 0 & 3 \\ 0 & \mathbf{5} \end{bmatrix} \text{ and } \begin{bmatrix} 5 \\ 2 \end{bmatrix} \text{ and } \begin{bmatrix} 6 \end{bmatrix} \text{ all have rank 1.}$$

For those matrices, the reduced row echelon $R =$ **rref** (A) can be checked by eye:

$$R = \begin{bmatrix} 1 & 3 & 4 \\ 0 & 0 & 0 \end{bmatrix} \text{ and } \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \text{ and } \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ and } \begin{bmatrix} 1 \end{bmatrix} \text{ have only one pivot.}$$

Our second definition of rank will be at a higher level. It deals with entire rows and entire columns—vectors and not just numbers. The matrices A and U and R have r *independent* rows (the pivot rows). They also have r *independent* columns (the pivot columns). Section 3.5 says what it means for rows or columns to be independent.

A third definition of rank, at the top level of linear algebra, will deal with *spaces* of vectors. ***The rank r is the "dimension" of the column space.*** It is also the dimension of the row space. The great thing is that r also reveals the dimension of the nullspace.

The Pivot Columns

The pivot columns of R have 1's in the pivots and 0's everywhere else. The r pivot columns taken together contain an r by r identity matrix I. It sits above $m - r$ rows of zeros. The numbers of the pivot columns are in the list *pivcol*.

The pivot columns of A are probably not obvious from A itself. But their column numbers are given by the *same list pivcol*. The r columns of A that eventually have pivots (in U and R) are the pivot columns of A. This example has $pivcol = (1, 3)$:

Pivot Columns

$$A = \begin{bmatrix} \mathbf{1} & 3 & \mathbf{0} & 2 & -1 \\ \mathbf{0} & 0 & \mathbf{1} & 4 & -3 \\ \mathbf{1} & 3 & \mathbf{1} & 6 & -4 \end{bmatrix} \text{ yields } R = \begin{bmatrix} \mathbf{1} & 3 & \mathbf{0} & 2 & -1 \\ \mathbf{0} & 0 & \mathbf{1} & 4 & -3 \\ \mathbf{0} & 0 & \mathbf{0} & 0 & 0 \end{bmatrix}.$$

The column spaces of A and R are different! All columns of this R end with zeros. Elimination subtracts rows 1 and 2 of A from row 3, to produce that zero row in R:

$$\boldsymbol{EA = R} \qquad \boldsymbol{A = E^{-1}R} \qquad E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix} \quad \text{and} \quad E^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}.$$

The r pivot columns of A are also the first r columns of E^{-1}. The r by r identity matrix inside R just picks out the first r columns of E^{-1} as columns of $A = E^{-1}R$.

One more fact about pivot columns. Their definition has been purely computational, based on R. Here is a direct mathematical description of the pivot columns of A:

The pivot columns are not combinations of earlier columns. The free columns ***are*** combinations of earlier columns. These combinations are the special solutions!

A pivot column of R (with 1 in the pivot row) cannot be a combination of earlier columns (with 0's in that row). The same column of A can't be a combination of earlier columns, because $Ax = \mathbf{0}$ *exactly when* $Rx = \mathbf{0}$.

Now we look at the special solution $\boldsymbol{x}$ from each free column.

The Special Solutions

Each special solution to $Ax = \mathbf{0}$ and $Rx = \mathbf{0}$ has one free variable equal to 1. The other free variables in $\boldsymbol{x}$ are all zero. The solutions come directly from the echelon form R:

Free columns
Free variables
in boldface

$$Rx = \begin{bmatrix} 1 & \mathbf{3} & 0 & \mathbf{2} & \mathbf{-1} \\ 0 & \mathbf{0} & 1 & \mathbf{4} & \mathbf{-3} \\ 0 & \mathbf{0} & 0 & \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Set the first free variable to $x_2 = 1$ with $x_4 = x_5 = 0$. The equations give the pivot variables $x_1 = -3$ and $x_3 = 0$. The special solution is $s_1 = (-3, 1, 0, 0, 0)$.

The next special solution has $x_4 = 1$. The other free variables are $x_2 = x_5 = 0$. The solution is $s_2 = (-2, 0, -4, 1, 0)$. Notice -2 and -4 in R, with plus signs.

The third special solution has $x_5 = 1$. With $x_2 = 0$ and $x_4 = 0$ we find $s_3 = (1, 0, 3, 0, 1)$. The numbers $x_1 = 1$ and $x_3 = 3$ are in column 5 of R, again with opposite signs. This is a general rule as we soon verify. The nullspace matrix N contains the three special solutions in its columns, so $AN = zero\ matrix$:

Nullspace matrix
$n - r = 5 - 2$
3 special solutions

$$N = \begin{bmatrix} -3 & -2 & 1 \\ 1 & 0 & 0 \\ 0 & -4 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{matrix} \text{not free} \\ \text{free} \\ \text{not free} \\ \text{free} \\ \text{free} \end{matrix}$$

The linear combinations of these three columns give all vectors in the nullspace. This is the complete solution to $Ax = 0$ (and $Rx = 0$). Where R had the identity matrix (2 by 2) in its pivot columns, N has the identity matrix (3 by 3) in its free rows.

There is a special solution for every free variable. Since r columns have pivots, that leaves $n - r$ free variables. This is the key to $Ax = 0$ and the nullspace:

$Ax = 0$ has r pivots and $n - r$ free variables: n columns minus r pivot columns. The ***nullspace matrix*** N contains the $n - r$ special solutions. Then $AN = 0$.

When we introduce the idea of "independent" vectors, we will show that the special solutions are independent. You can see in N that no column is a combination of the other columns. The beautiful thing is that the count is exactly right:

$Ax = 0$ has r independent equations so it has $\mathbf{n - r}$ independent solutions.

The special solutions are easy for $Rx = 0$. Suppose that the first r columns are the pivot columns. Then the reduced row echelon form looks like

$$R = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix} \begin{matrix} r \text{ pivot rows} \\ m - r \text{ zero rows} \end{matrix} \qquad (4)$$

r pivot columns $\quad n - r$ free columns

The pivot variables in the $n - r$ special solutions come by changing F to $-F$:

$$\textbf{Nullspace matrix} \quad N = \begin{bmatrix} -F \\ I \end{bmatrix} \begin{matrix} r \text{ pivot variables} \\ n - r \text{ free variables} \end{matrix} \qquad (5)$$

Check $RN = 0$. The first block row of RN is (I times $-F$) + (F times I) = zero. The columns of N solve $Rx = 0$. When the free part of $Rx = 0$ moves to the right side,

the left side just holds the identity matrix:

$$Rx = 0 \quad \text{means} \quad I\begin{bmatrix} \text{pivot} \\ \text{variables} \end{bmatrix} = -F\begin{bmatrix} \text{free} \\ \text{variables} \end{bmatrix}. \tag{6}$$

In each special solution, the free variables are a column of I. Then the pivot variables are a column of $-F$. Those special solutions give the nullspace matrix N.

The idea is still true if the pivot columns are mixed in with the free columns. Then I and F are mixed together. You can still see $-F$ in the solutions. Here is an example where $I = [1]$ comes first and $F = \begin{bmatrix} 2 & 3 \end{bmatrix}$ comes last.

Example 2 The special solutions of $Rx = x_1 + 2x_2 + 3x_3 = 0$ are the columns of N:

$$R = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \qquad N = \begin{bmatrix} -F \\ I \end{bmatrix} = \begin{bmatrix} -2 & -3 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The rank is one. There are $n - r = 3 - 1$ special solutions $(-2, 1, 0)$ and $(-3, 0, 1)$.

Final Note How can I write confidently about R not knowing which steps MATLAB will take? A could be reduced to R in different ways. Very likely you and Mathematica and Maple would do the elimination differently. The key is that ***the final R is always the same.*** *The original A completely determines the I and F and zero rows in R.*

For proof I will determine the pivot columns (which locate I) and free columns (which contain F) in an "algebra way"—two rules that have nothing to do with any particular elimination steps. Here are those rules:

1. The pivot columns *are not* combinations of earlier columns of A.
2. The free columns *are* combinations of earlier columns (F tells the combinations).

A small example with rank one will show two E's that produce the correct $EA = R$:

$$A = \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix} \text{ reduces to } R = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} = \text{rref}(A) \quad \text{and no other } R.$$

You could multiply row 1 of A by $\frac{1}{2}$, and subtract row 1 from row 2:

Two steps give E
$$\begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 1/2 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1/2 & 0 \\ -1/2 & 1 \end{bmatrix} = E.$$

Or you could exchange rows in A, and then subtract 2 times row 1 from row 2:

Two different steps give E_{new}
$$\begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix} = E_{\text{new}}.$$

Multiplication gives $EA = R$ and also $E_{\text{new}}A = R$. ***Different E's but the same R.***

Codes for Row Reduction

There is no way that rref will ever come close in importance to lu. The Teaching Code elim for this book uses rref. Of course rref(R) would give R again!

MATLAB: $[R, pivcol] = \text{rref}(A)$ Teaching Code: $[E, R] = \text{elim}(A)$

The extra output *pivcol* gives the numbers of the pivot columns. They are the same in A and R. The extra output E in the Teaching Code is an m by m ***elimination matrix*** that puts the original A (whatever it was) into its row reduced form R:

$$EA = R.$$

The square matrix E is the product of elementary matrices E_{ij} and also P_{ij} and D^{-1}. P_{ij} exchanges rows. The diagonal D^{-1} divides rows by their pivots to produce 1's.

If we want E, we can apply row reduction to the matrix $\begin{bmatrix} A & I \end{bmatrix}$ with $n + m$ columns. All the elementary matrices that multiply A (to produce R) will also multiply I (to produce E). The whole augmented matrix is being multiplied by E:

$$E \begin{bmatrix} A & I \end{bmatrix} = \begin{bmatrix} R & E \end{bmatrix} \tag{7}$$

This is exactly what "Gauss-Jordan" did in Chapter 2 to compute A^{-1}. ***When A is square and invertible, its reduced row echelon form is I.*** Then $EA = R$ becomes $EA = I$. In this invertible case, E is A^{-1}. This chapter is going further, to every A.

■ REVIEW OF THE KEY IDEAS ■

1. The rank r of A is the number of pivots (which are 1's in $R = \text{rref(A)}$).
2. The r pivot columns of A and R are in the same list *pivcol*.
3. Those r pivot columns are not combinations of earlier columns.
4. The $n - r$ free columns *are* combinations of earlier columns (pivot columns).
5. Those combinations (using $-F$ taken from R) give the $n - r$ special solutions to $Ax = \mathbf{0}$ and $Rx = \mathbf{0}$. They are the $n - r$ columns of the nullspace matrix N.

■ WORKED EXAMPLES ■

3.3 A Find the reduced echelon form of A. What is the rank? What is the special solution to $Ax = \mathbf{0}$?

Second differences $-1, 2, -1$
Notice $A_{11} = A_{44} = 1$

$$A = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

Solution Add row 1 to row 2. Then add row 2 to row 3. Then add row 3 to row 4:

First differences 1, −1
$$U = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Now add row 3 to row 2. Then add row 2 to row 1:

Reduced form
$$R = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}.$$

The rank is $r = 3$. There is one free variable ($n - r = 1$). *The special solution is* $s = (1, 1, 1, 1)$. *Every row adds to* 0. Notice $-F = (1, 1, 1)$ in the pivot variables of s.

3.3 B Factor these rank one matrices into $A = uv^{\mathrm{T}}$ = column times row:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix} \qquad A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \text{(find } d \text{ from } a, b, c \text{ if } a \neq 0)$$

Split this rank two matrix into $u_1v_1^{\mathrm{T}} + u_2v_2^{\mathrm{T}}$ = (3 by 2) times (2 by 4) using R:

$$A = \begin{bmatrix} 1 & 1 & 0 & 2 \\ 1 & 2 & 0 & 3 \\ 2 & 3 & 0 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 0 \\ 2 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = E^{-1}R.$$

Solution For the 3 by 3 matrix A, all rows are multiples of $v^{\mathrm{T}} = [1\ \ 2\ \ 3]$. All columns are multiples of the column $u = (1, 2, 3)$. This symmetric matrix has $u = v$ and A is uu^{T}. Every rank one symmetric matrix will have this form or else $-uu^{\mathrm{T}}$.

If the 2 by 2 matrix $\left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$ has rank one, it must be singular. In Chapter 5, its determinant is $ad - bc = 0$. In this chapter, row 2 is c/a times row 1.

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 \\ c/a \end{bmatrix} \begin{bmatrix} a & b \end{bmatrix} = \begin{bmatrix} a & b \\ c & bc/a \end{bmatrix}. \quad \text{So} \quad d = \frac{bc}{a}.$$

The 3 by 4 matrix of rank two is a sum of *two matrices of rank one.* All columns of A are combinations of the pivot columns 1 and 2. All rows are combinations of the nonzero rows of R. The pivot columns are u_1 and u_2 and those rows are v_1^{T} and v_2^{T}. Then A is $u_1v_1^{\mathrm{T}} + u_2v_2^{\mathrm{T}}$, multiplying r columns of E^{-1} times r rows of R:

Columns times rows
$$\begin{bmatrix} 1 & 1 & 0 & 2 \\ 1 & 2 & 0 & 3 \\ 2 & 3 & 0 & 5 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 1 \end{bmatrix}$$

3.3 C Find the row reduced form R and the rank r of A and B (*those depend on* c). Which are the pivot columns of A? What are the special solutions and the matrix N?

Find special solutions $A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 6 & 3 \\ 4 & 8 & c \end{bmatrix}$ and $B = \begin{bmatrix} c & c \\ c & c \end{bmatrix}$.

Solution The matrix A has rank $r = 2$ ***except if*** $c = 4$. The pivots are in columns 1 and 3. The second variable x_2 is free. Notice the form of R:

$$c \neq 4 \quad R = \begin{bmatrix} \mathbf{1} & 2 & 0 \\ 0 & 0 & \mathbf{1} \\ 0 & 0 & 0 \end{bmatrix} \qquad c = 4 \quad R = \begin{bmatrix} \mathbf{1} & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Two pivots leave one free variable x_2. But when $c = 4$, the only pivot is in column 1 (rank one). The second and third variables are free, producing two special solutions:

$$c \neq 4 \quad \text{Special solution with } x_2 = 1 \text{ goes into } N = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}.$$

$$c = 4 \quad \text{Another special solution goes into } N = \begin{bmatrix} -2 & -1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The 2 by 2 matrix $\begin{bmatrix} c & c \\ c & c \end{bmatrix}$ has rank $r = 1$ ***except if*** $c = 0$, when the rank is zero!

$$c \neq 0 \quad R = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \text{ and } N = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad \text{Nullspace} = \text{line}$$

The matrix has *no pivot columns* if $c = 0$. Then both variables are free:

$$c = 0 \quad R = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \text{ and } N = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{Nullspace} = \mathbf{R}^2.$$

Problem Set 3.3

1 Which of these rules gives a correct definition of the *rank* of A?

(a) The number of nonzero rows in R.

(b) The number of columns minus the total number of rows.

(c) The number of columns minus the number of free columns.

(d) The number of 1's in the matrix R.

2 Find the reduced row echelon forms R and the rank of these matrices:

(a) The 3 by 4 matrix with all entries equal to 4.

(b) The 3 by 4 matrix with $a_{ij} = i + j - 1$.

(c) The 3 by 4 matrix with $a_{ij} = (-1)^j$.

3 Find the reduced R for each of these (block) matrices:

$$A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 3 \\ 2 & 4 & 6 \end{bmatrix} \quad B = \begin{bmatrix} A & A \end{bmatrix} \quad C = \begin{bmatrix} A & A \\ A & 0 \end{bmatrix}$$

4 Suppose all the pivot variables come *last* instead of first. Describe all four blocks in the reduced echelon form (the block B should be r by r):

$$R = \begin{bmatrix} A & B \\ C & D \end{bmatrix}.$$

What is the nullspace matrix N containing the special solutions?

5 (Silly problem) Describe all 2 by 3 matrices A_1 and A_2, with row echelon forms R_1 and R_2, such that $R_1 + R_2$ is the row echelon form of $A_1 + A_2$. Is is true that $R_1 = A_1$ and $R_2 = A_2$ in this case? Does $R_1 - R_2$ equal $\mathbf{rref}(A_1 - A_2)$?

6 If A has r pivot columns, how do you know that A^{T} has r pivot columns? Give a 3 by 3 example with different column numbers in *pivcol* for A and A^{T}.

7 What are the special solutions to $R\boldsymbol{x} = \mathbf{0}$ and $\boldsymbol{y}^{\mathrm{T}}R = \mathbf{0}$ for these R?

$$R = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 4 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad R = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Problems 8–11 are about matrices of rank $r = 1$.

8 Fill out these matrices so that they have rank 1:

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & & \\ 4 & & \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} & 9 & \\ 1 & & \\ 2 & 6 & -3 \end{bmatrix} \quad \text{and} \quad M = \begin{bmatrix} a & b \\ c & \end{bmatrix}.$$

9 If A is an m by n matrix with $r = 1$, its columns are multiples of one column and its rows are multiples of one row. The column space is a _____ in $\mathbf{R}^m$. The nullspace is a _____ in $\mathbf{R}^n$. The nullspace matrix N has shape _____.

10 Choose vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ so that $A = \boldsymbol{u}\boldsymbol{v}^{\mathrm{T}}$ = column times row:

$$A = \begin{bmatrix} 3 & 6 & 6 \\ 1 & 2 & 2 \\ 4 & 8 & 8 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 2 & 2 & 6 & 4 \\ -1 & -1 & -3 & -2 \end{bmatrix}.$$

$A = \boldsymbol{u}\boldsymbol{v}^{\mathrm{T}}$ is the natural form for every matrix that has rank $r = 1$.

11 If A is a rank one matrix, the second row of U is _____. Do an example.

Problems 12–14 are about r by r invertible matrices inside A.

12 *If A has rank r, then it has an r by r submatrix S that is invertible.* Remove $m-r$ rows and $n-r$ columns to find an invertible submatrix S inside A, B, and C. You could keep the pivot rows and pivot columns:

$$A=\begin{bmatrix}1&2&3\\1&2&4\end{bmatrix} \qquad B=\begin{bmatrix}1&2&3\\2&4&6\end{bmatrix} \qquad C=\begin{bmatrix}0&1&0\\0&0&0\\0&0&1\end{bmatrix}.$$

13 Suppose P contains only the r pivot columns of an m by n matrix. Explain why this m by r submatrix P has rank r.

14 Transpose P in problem 13. Then find the r pivot columns of P^{T}. Transposing back, **this produces an r by r invertible submatrix S inside P and A:**

$$\text{For } A=\begin{bmatrix}1&2&3\\2&4&6\\2&4&7\end{bmatrix} \text{ find } P \text{ (3 by 2) and then the invertible } S \text{ (2 by 2).}$$

Problems 15–20 show that rank(AB) is not greater than rank(A) or rank(B).

15 Find the ranks of AB and AC (rank one matrix times rank one matrix):

$$A=\begin{bmatrix}1&2\\2&4\end{bmatrix} \text{ and } B=\begin{bmatrix}2&1&4\\3&1.5&6\end{bmatrix} \text{ and } C=\begin{bmatrix}1&b\\c&bc\end{bmatrix}.$$

16 The rank one matrix $\boldsymbol{uv}^{\mathrm{T}}$ times the rank one matrix $\boldsymbol{wz}^{\mathrm{T}}$ is $\boldsymbol{uz}^{\mathrm{T}}$ times the number _____. This product $\boldsymbol{uv}^{\mathrm{T}}\boldsymbol{wz}^{\mathrm{T}}$ also has rank one unless _____ $= 0$.

17 (a) Suppose column j of B is a combination of previous columns of B. Show that column j of AB is the same combination of previous columns of AB. Then AB cannot have new pivot columns, so **rank(AB) $\leq$ rank(B).**

(b) Find A_1 and A_2 so that $\text{rank}(A_1B)=1$ and $\text{rank}(A_2B)=0$ for $B=\begin{bmatrix}1&1\\1&1\end{bmatrix}$.

18 Problem 17 proved that $\text{rank}(AB) \leq \text{rank}(B)$. Then the same reasoning gives **rank($B^{\mathrm{T}}A^{\mathrm{T}}$) $\leq$ rank(A^{T}).** How do you deduce that **rank(AB) $\leq$ rank A?**

19 *(Important)* Suppose A and B are n by n matrices, and $AB = I$. Prove from $\text{rank}(AB) \leq \text{rank}(A)$ that the rank of A is n. So A is invertible and B must be its two-sided inverse (Section 2.5). Therefore $BA = I$ *(which is not so obvious!).*

20 If A is 2 by 3 and B is 3 by 2 and $AB = I$, show from its rank that $BA \neq I$. Give an example of A and B with $AB = I$. For $m < n$, a right inverse is not a left inverse.

21 Suppose A and B have the *same* reduced row echelon form R.

(a) Show that A and B have the same nullspace and the same row space.

(b) We know $E_1A = R$ and $E_2B = R$. So A equals an _____ matrix times B.

22 Express A and then B as the sum of two rank one matrices:

$$\textbf{rank} = 2 \qquad A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 4 \\ 1 & 1 & 8 \end{bmatrix} \qquad B = \begin{bmatrix} 2 & 2 \\ 2 & 3 \end{bmatrix}.$$

23 Answer the same questions as in Worked Example **3.3 C** for

$$A = \begin{bmatrix} 1 & 1 & 2 & 2 \\ 2 & 2 & 4 & 4 \\ 1 & c & 2 & 2 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1-c & 2 \\ 0 & 2-c \end{bmatrix}.$$

24 What is the nullspace matrix N (containing the special solutions) for A, B, C?

$$A = \begin{bmatrix} I & I \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} I & I \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} I & I & I \end{bmatrix}.$$

25 *Neat fact* ***Every m by n matrix of rank r reduces to*** (m by r) ***times*** (r by n):

$$A = (\text{pivot columns of } A)(\text{first } r \text{ rows of } R) = (\textbf{COL})(\textbf{ROW}).$$

Write the 3 by 4 matrix A in equation (1) at the start of this section as the product of the 3 by 2 matrix from the pivot columns and the 2 by 4 matrix from R.

Challenge Problems

26 Suppose A is an m by n matrix of rank r. Its reduced echelon form is R. Describe exactly the matrix Z (its shape and all its entries) that comes from *transposing the reduced row echelon form of* R' (prime means transpose):

$$R = \text{rref}(A) \quad \text{and} \quad Z = (\text{rref}(R'))'.$$

27 Suppose R is m by n of rank r, with pivot columns first:

$$R = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}.$$

(a) What are the shapes of those four blocks?

(b) Find a *right-inverse* B with $RB = I$ if $r = m$.

(c) Find a *left-inverse* C with $CR = I$ if $r = n$.

(d) What is the reduced row echelon form of R^{T} (with shapes)?

(e) What is the reduced row echelon form of $R^{\mathrm{T}}R$ (with shapes)?

Prove that $R^{\mathrm{T}}R$ has the same nullspace as R. Later we show that $A^{\mathrm{T}}A$ always has the same nullspace as A (a valuable fact).

28 Suppose you allow elementary *column* operations on A as well as elementary row operations (which get to R). What is the "row-and-column reduced form" for an m by n matrix of rank r?